

How Is LLM Reasoning Distracted by Irrelevant Context?

An Analysis Using a Controlled Benchmark

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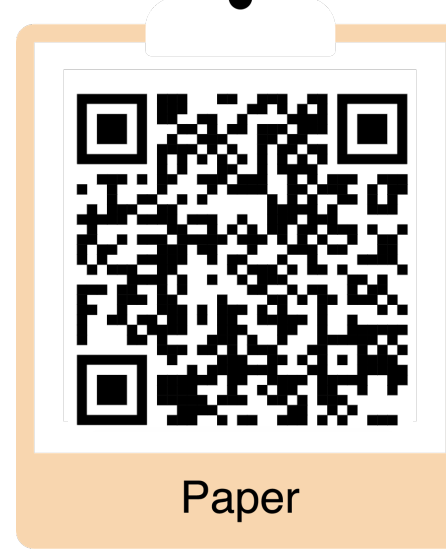
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Introduction

How robust is LLM reasoning?

LLM is easy to be affected by **irrelevant context**.

Flanker Effect:

When a target stimulus is surrounded by distractors suggesting a different response, people take longer to respond and tend to make more mistakes.

Challenges:

- 1) How does varying the amount of irrelevant context affect robustness?
- 2) Can robust reasoning be enhanced through continued pretraining or LoRA?
- 3) How does the intensity of irrelevant context during training impact model performance in both in-distribution and OOD scenarios?
- 4) How can the above questions be qualitatively evaluated?

Solutions:

GSM-DC - A synthetic benchmark

- The explicit injection of irrelevant context via off-path nodes and edges without affecting correct solutions.
- Adjustment of reasoning complexity by varying graph depth and structure.
- Automatic evaluation of model outputs.
- Exploration through controlled experiments.

Metrics

Automatic stepwise evaluation of solutions by comparing with the correct reasoning path:

Step Accuracy (SAcc):

- Each step must compute the correct value using only reachable nodes in G' .
- Extra steps are allowed if they don't interfere.

Path Accuracy (PAcc):

- The predicted reasoning must node-level aligned with solution path P .
- Permitting redundancy but not confused by irrelevant context.

Extraction Answer Accuracy (EAcc):

- The final answer must match the answer from ground-truth solution S .

Note: All metrics are computed using a symbolic parser with node-level alignment, not strict sentence-level sequence matching.

Limitations

Broader applicability:

- Methodology applies to any symbolic reasoning task (e.g., logic, algorithms).

Extension to non-unique reasoning paths:

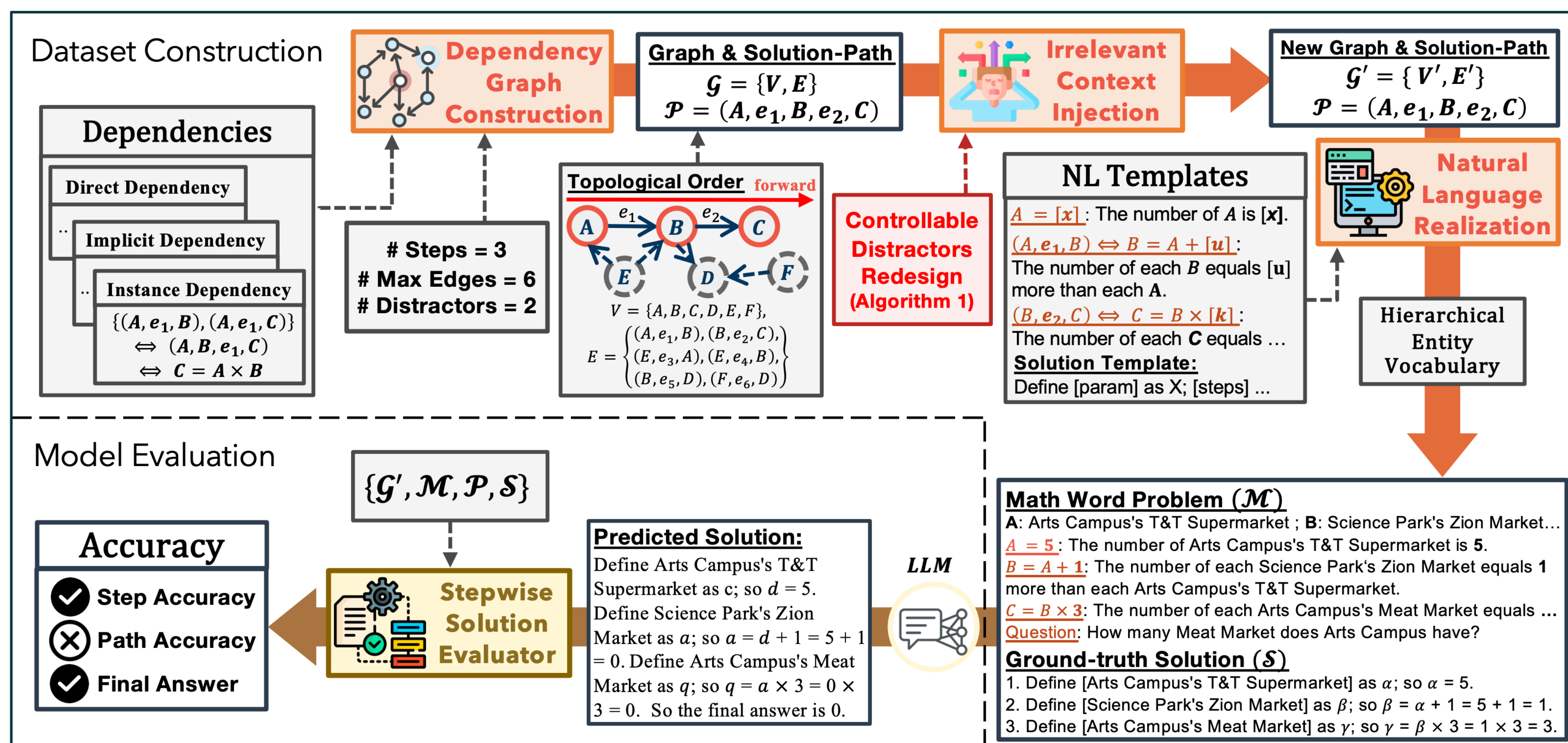
- Allow multiple valid reasoning chains.

Plans for new evaluations:

- RL-based training using Process Reward Models.
- Designing stepwise evaluator to evaluate reasoning models such as OpenAI o1/o3/o4 and DeepSeek-R1.

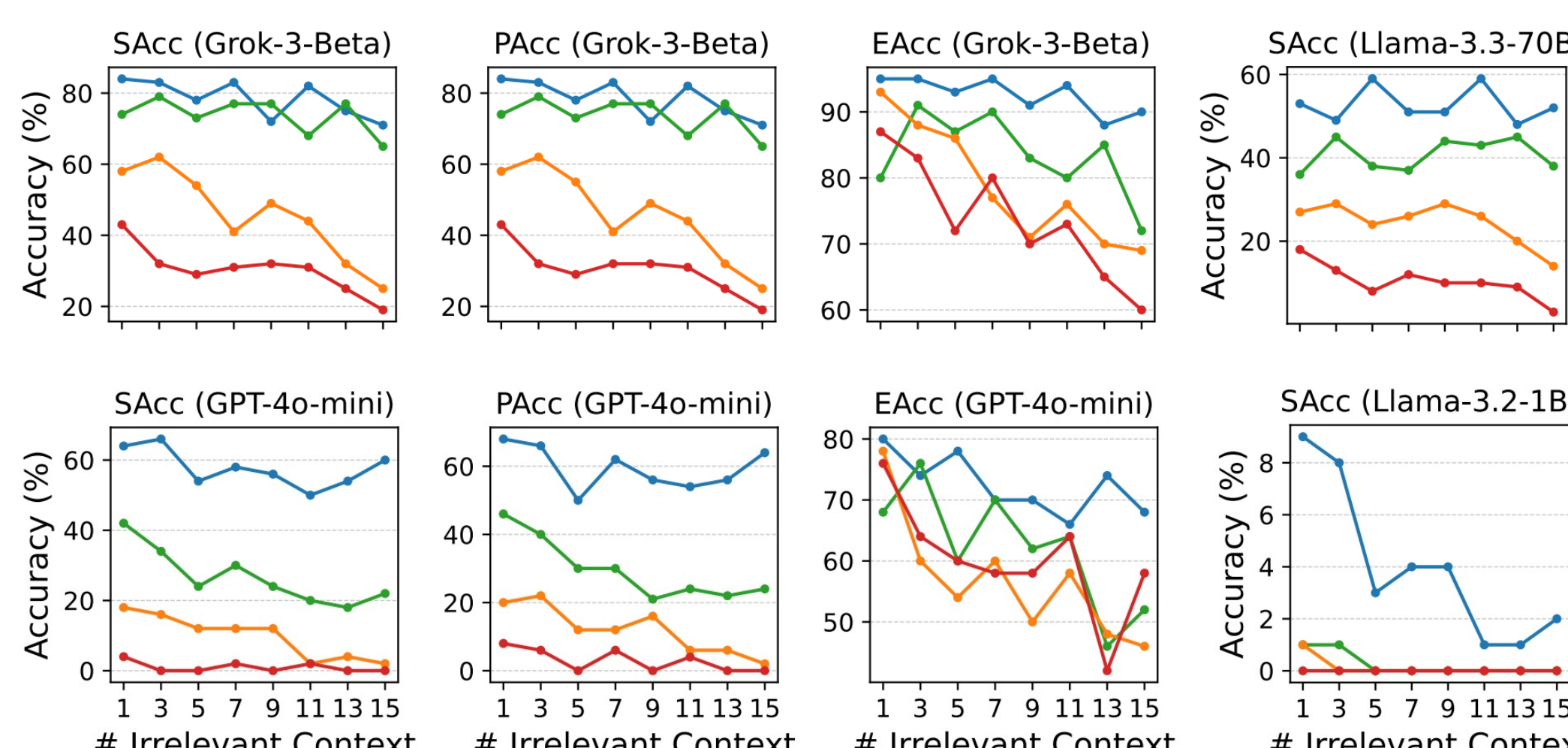
Graph-Based Benchmark for Controlled Experiments

General Framework (Gen&Eval): Grade School Math with Distracting Context (GSM-DC)

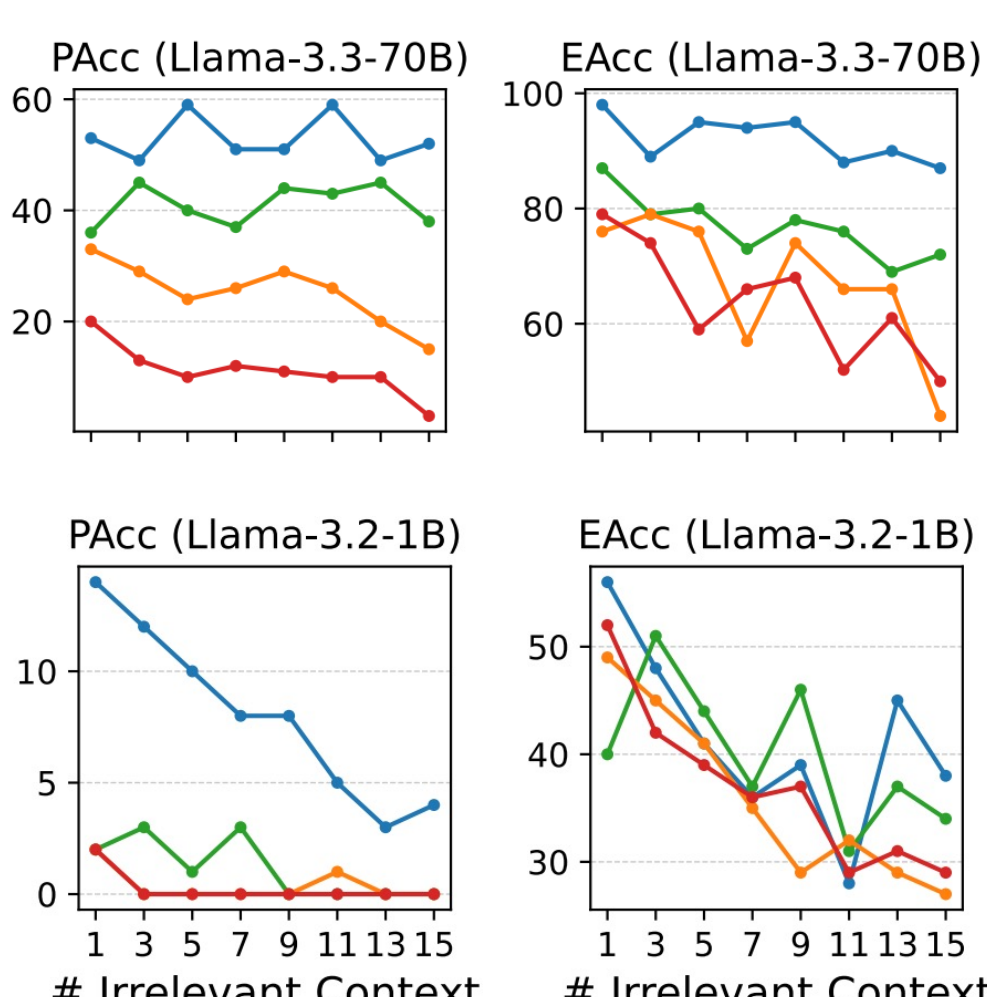


Results from Controlled Experiments

Result 1: LLMs' reasoning performance degrades with increasing irrelevant context.



Result 2: Irrelevant context degrades accuracy more steeply at greater reasoning depths.



Distractor Scaling Law

$$E(m; rs) = m^{\delta(rs)}$$

- $E(m; rs)$: error rate
- m : distractor count
- rs : reasoning steps
- $\delta(rs)$: model's sensitivity to distractors at reasoning steps rs

Result 3: Continued pretraining enhances robustness even without access to IC samples.

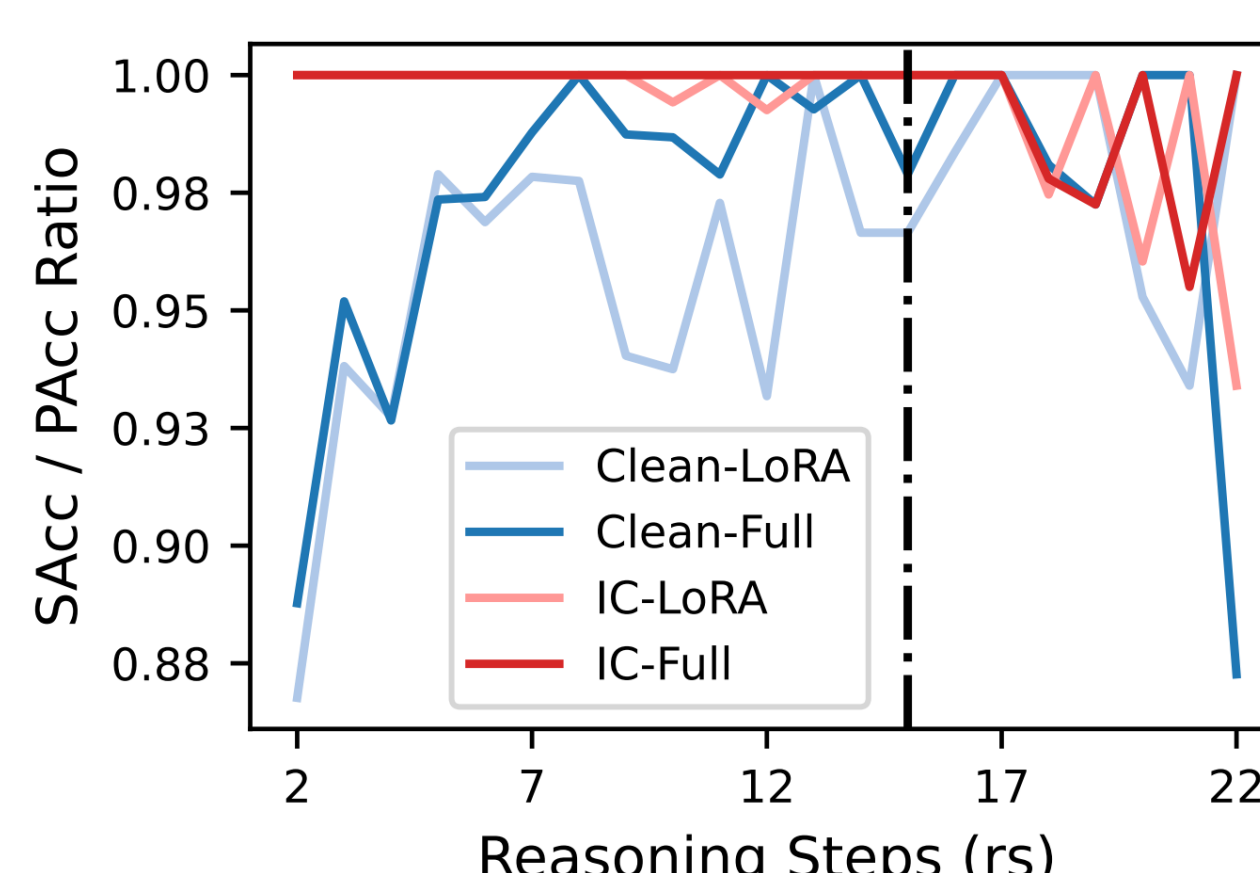


Fig: Step accuracy of models trained with Non-IC or IC data using LoRA or continued pretraining.

Result 4: Training with irrelevant context improves robustness most effectively.

rs	Clean		Clean+IC		IC	
	SAcc	PAcc	SAcc	PAcc	SAcc	PAcc
≤ 15	35.9	41.3	70.0	71.2	73.2	74.7
16	22.0	22.7	32.0	32.0	33.3	33.3
17	21.0	21.0	23.0	23.0	20.7	21.3
18	13.0	13.0	15.7	15.7	16.7	16.7
19	13.7	13.7	13.3	13.3	15.0	15.0
20	9.0	9.0	8.3	8.3	10.0	10.0
21	7.7	7.7	8.7	8.7	5.7	5.7
22	6.0	6.0	5.3	5.3	6.3	6.3

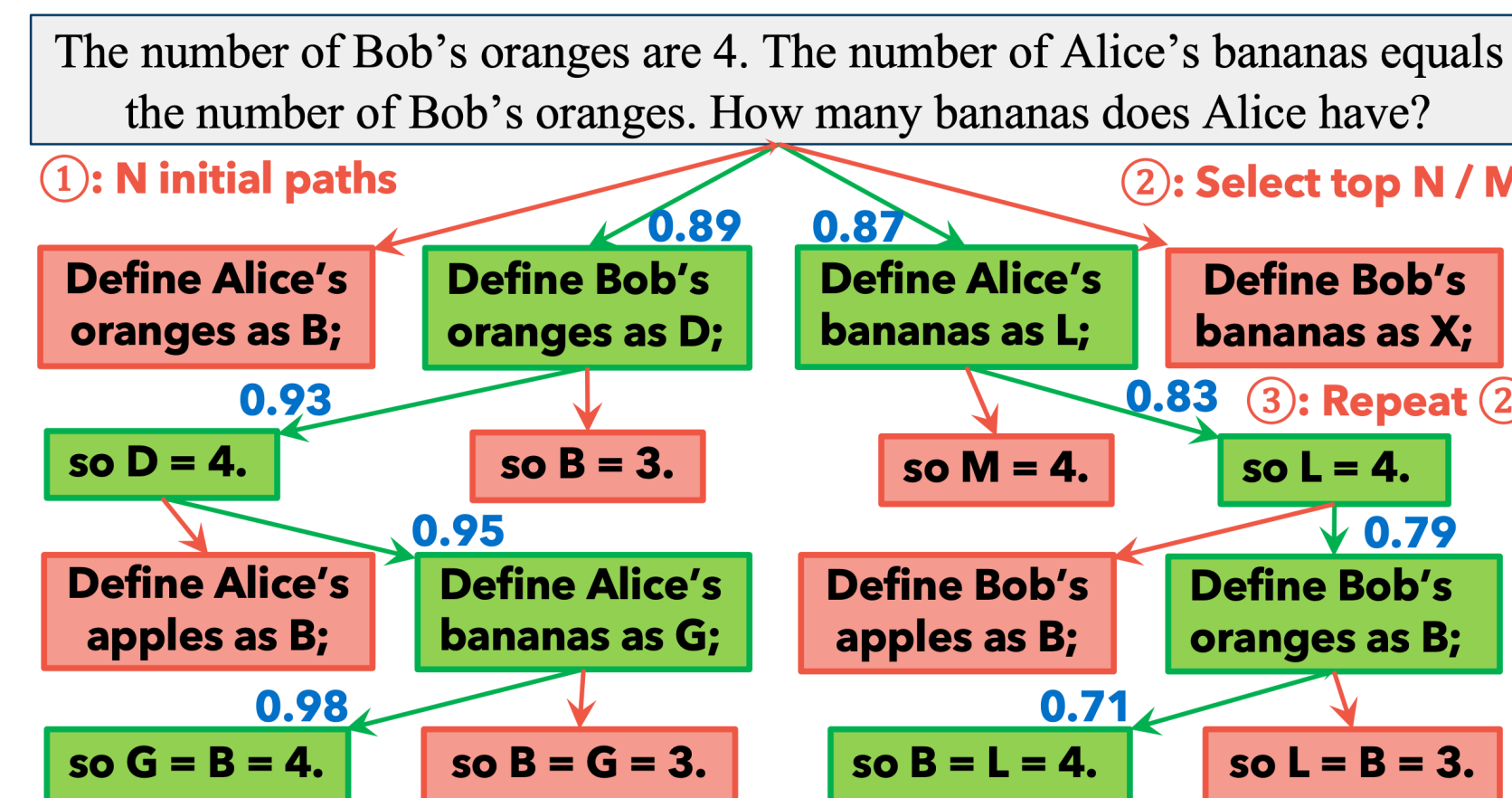
Fig: Comparison of SAcc and PAcc under different training regimes: Clean, Clean+IC, and IC.

Result 5: Training with challenging irrelevant context leads to the strongest robustness and generalization across all pretraining settings.

Training IC Level	Testing w/ IC (SAcc)			Testing w/o IC (SAcc)		
	ID	OOD	All	ID	OOD	All
CLEAN	35.91	13.19	32.36	81.95	17.05	60.32
LIGHT-IC	64.79	6.90	46.57	67.33	7.09	46.56
MEDIUM-IC	65.79	7.23	47.44	69.39	9.95	50.38
HARD-IC	77.95	18.57	59.48	82.30	19.86	61.21
MIX-IC	73.23	15.33	57.86	78.09	15.62	57.38

Training IC Level	ID Test SAcc			OOD Test SAcc		
	Light	Medium	Hard	Light	Medium	Hard
LIGHT-IC	67.21	66.57	60.57	8.14	7.29	5.28
MEDIUM-IC	68.14	66.07	63.14	8.71	8.43	4.57
HARD-IC	78.36	79.21	76.28	22.7	18.43	14.57
MIX-IC	74.71	75.07	69.93	17.7	16.57	11.28

Result 6: Improving reasoning robustness at test time: Tree search can enhance the generalization capabilities of LLMs.



Training IC Level	ID SAcc			OOD SAcc		
	w/o PRM	w/ PRM	Δ	w/o PRM	w/ PRM	Δ
LIGHT-IC	64.79	66.10	+1.31	6.90	9.59	+2.69
MEDIUM-IC	65.79	70.05	+4.26	7.23	13.52	+6.29
HARD-IC	77.95	79.48	+1.53	18.57	24.17	+5.60
MIX-IC	73.23	75.81	+2.58	15.33	19.06	+3.73
CLEAN	35.91	36.38	+0.47	13.19	15.76	+2.57